

Multi-frequency Crystal Core Scale Prevention System

Industrial physical water treatment equipment · Промышленное оборудование физической водоподготовки

PHYSICAL TREATMENT	In-line treatment without chemical agents. No external power is required.
WORKING PRINCIPLE	Water passes through the crystal core, impeller and multi-frequency disturbance field. Preset natural-frequency information in the core produces micro-vibrations acting on water and mineral molecules, reducing scale deposition. Existing scale is described as gradually loosening into a sand-like state.
SERVICE	The supplied documentation states no routine personnel attendance or maintenance and a general service life of at least 10 years, depending on operating conditions.

FR Series Technical Specifications

Standard range from DN25 to DN350. Values below are reproduced from the supplied company materials.

Model	Pipe DN (mm)	Max. D (mm)	Height H (mm)	Flow (m³/h)	Connection	Pressure (MPa)
FR-DN25	DN25	45	222	0.01-7.20	Thread / Flange	1.0 / 1.6
FR-DN40	DN40	80	259	7.21-13.70	Thread / Flange	1.0 / 1.6
FR-DN80	DN80	120	299	13.71-41.40	Thread / Flange	1.0 / 1.6
FR-DN100	DN100	220	386	41.41-50.30	Flange	1.0 / 1.6 / 2.5
FR-DN150	DN150	285	455	50.31-124.90	Flange	1.0 / 1.6 / 2.5
FR-DN200	DN200	340	430	124.91-242.30	Flange	1.0 / 1.6 / 2.5
FR-DN250	DN250	405	532	242.31-389.80	Flange	1.0 / 1.6 / 2.5
FR-DN300	DN300	460	488	389.81-539.90	Flange	1.0 / 1.6 / 2.5
FR-DN350	DN350	520	556	539.91-667.90	Flange	1.0 / 1.6 / 2.5

Note: Dimensions shown are for GB-standard 1.0 MPa flanges. Dimensions are adjusted when 1.6 MPa or 2.5 MPa flanges are selected. Other sizes may be customized for the application.

Selection and Application

Selection

Match the device model to the pipe diameter at the installation point. Project discussion should use the actual pipe size, flow, pressure and water conditions.

Installation point

Install at the protected equipment inlet or after the shut-off valve at the pump outlet. Align the arrow on the body with the direction of flow.

Circulating water

Install on return-water and make-up-water lines. Treat make-up water before it enters the circuit.

Temperature and water

Treat ambient-temperature water before heating. The supplied material states that water quality is not restricted; special water requires customized selection.

Application fields

Power and heat, boiler water, cooling circulation, hot-water exchange, pipelines, beverage, petrochemical, food, metallurgy and mining, and pharmaceutical applications.

Application conditions · Условия применения

Standard range	Water temperature	Flow velocity	Pressure options
DN25-DN350	5-85°C	0.5-2.0 m/s	1.0 / 1.6 / 2.5 MPa

This manual summarizes only the information supplied by the company. Confirm model selection against actual project conditions.